Autonomous Scarecrow Robot

HND in Computer Science with Artificial Intelligence

Project – Proposal

HNDCSAI-261-F

School of Computing and Engineering

National Institute of Business Management

Sri Lanka

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT (NIBM)

HND in Computer Science with Artificial Intelligence

IOT & ROBOTICS PROJECT PROPOSAL

Project Title: Autonomous Scarecrow Robot

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Chapter 1

Introduction

1.1 Identifying the Purpose

Gardens and small farms are often affected by animals, birds, and unauthorized human activities. Traditional scarecrows are commonly used to scare birds away from crops. However, traditional scarecrows are static and ineffective over time, as birds and animals eventually become familiar with them.

Therefore, there is a need for a smart and automated solution that can actively monitor the garden area and detect movement.

This project proposes an Autonomous Scarecrow Robot that patrols the garden using a line-following system. The robot stops at specific locations and scans the surroundings using a PIR motion sensor. When movement is detected, the robot activates an alarm sound and actuators to scare away intruders.

This system improves garden protection, automation, and monitoring efficiency.

1.2 Identifying Specific Requirements

The robot must:

- Move autonomously using a line-following system
- Stop at predefined locations
- Perform 360-degree scanning
- Detect motion using a PIR sensor
- Trigger alarm sound using a speaker
- Activate actuators
- Continue patrolling after scanning
- Operate using battery power

MAIN FUNCTIONS

1. Autonomous Navigation Function

What it does:

- The robot moves along a predefined path using line-following sensors.

Purpose:

- Allows the robot to patrol the garden automatically without manual control.

2. Stop Point Monitoring Function

What it does:

- The robot stops at predefined locations in the garden.

Purpose:

- Allows the robot to monitor important areas such as crop sections or entry points.

3. Motion Detection Function (PIR Sensor)

What it does:

- Detects movement of humans or animals using a PIR motion sensor.

Purpose:

- Helps identify intruders or animals entering the garden.

4. 360° Scanning Function

What it does:

- A servo motor rotates the PIR sensor to scan the environment at 90-degree intervals, completing a full 360-degree scan.

Purpose:

- Allow the robot to monitor the entire surrounding area.

5. Alarm Function

What it does:

- Activates a buzzer when motion is detected.

Purpose:

- Produces sound to scare away animals or intruders.

6. Actuator Activation Function

What it does:

- Activate mechanical movements to scare away detected objects.

Purpose:

- Improves the effectiveness of the scarecrow system.

7. Obstacle Detection Function

What it does:

- The ultrasonic sensor measures the distance between the robot and objects in front of it. When an obstacle is detected within a predefined distance, the robot stops automatically.

Purpose:

- Prevents the robot from colliding with objects such as plants, animals, or other obstacles in the garden. It also ensures safe and smooth navigation.

Chapter 2

Researching and Designing

2.1 Gathering Information and Feasibility Study

Before developing the robotic scarecrow system, research was conducted on the technologies and components required for the project. The following areas were studied:

Research areas:

- Line-following robot technology
- PIR motion sensors
- Servo motor scanning systems
- Arduino microcontroller programming
- Robotic security systems

The research helped identify suitable components and methods that could be used to implement the system.

Feasibility:

The feasibility of the proposed system was evaluated based on technical and economic factors.

- Arduino microcontroller provides an easy and flexible platform for controlling sensors and motors.
- PIR sensors provide reliable motion detection.
- Line-following sensors can help the robot navigate along a predefined path.
- L298N motor driver module can control DC motors that move the robot.
- A servo motor can be used to rotate the PIR sensor to perform a full environmental scan.
- All components required for the project are low cost and easily available in local electronic stores.

Based on this study, it was concluded that the proposed robotic scarecrow system is technically feasible and economically affordable using commonly available electronic components.

2.2 Design Requirements

During the design phase, several important details were considered:

- The robot must have a stable chassis structure to support motors and electronic components.
- The line-following sensors must be accurately positioned to detect the path.
- The PIR sensor must be mounted on a rotating servo motor to perform 360-degree scanning.
- The system must include an alarm mechanism such as a buzzer or speaker.
- The robot must be powered by a reliable battery supply.
- The robot must detect and avoid obstacles using an ultrasonic sensor.

2.3 Possible and Alternative Design Solutions

Several design approaches were considered before selecting the final design:

Alternative Design:

- **Fixed Scarecrow with Sensors**

  A stationary scarecrow equipped with motion sensors could detect movement and trigger alarms. However, this design cannot monitor the entire garden area.
- **Camera-Based Monitoring System**

  A camera-based surveillance system could monitor the garden continuously. However, this approach requires more complex image processing and higher cost.

Selected Design:

- **Mobile Robotic Scarecrow**

  The selected solution is a mobile robotic scarecrow that follows a path around the garden.

This design allows the robot to monitor multiple locations while keeping the system simple and cost-effective.

2.4 Planning and Structural Design

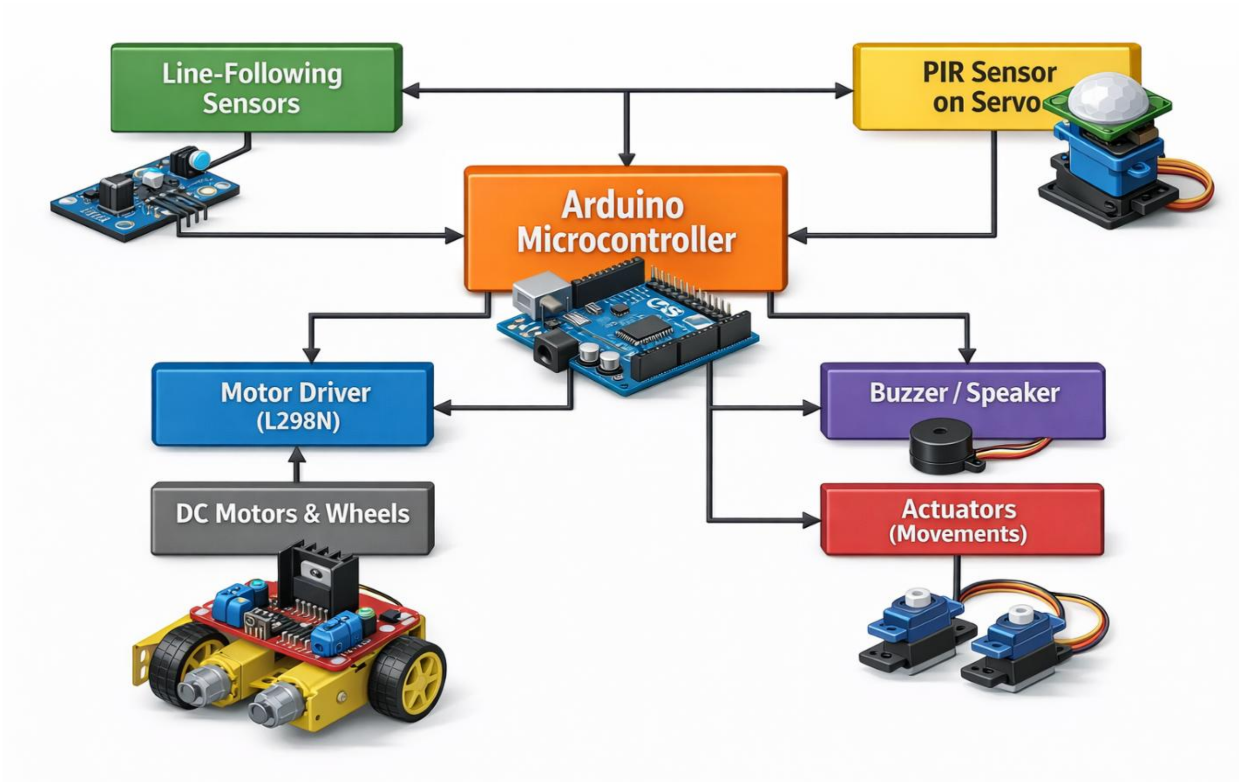
The robot structure will consist of:

- A robot chassis with wheels and DC motors
- Line-following sensors mounted at the front
- An ultrasonic sensor mounted at the front for obstacle detection
- A PIR sensor mounted on a rotating servo motor
- A buzzer or speaker for sound alerts
- Actuators to create movement for scary intruders
- An Arduino microcontroller to control the entire system

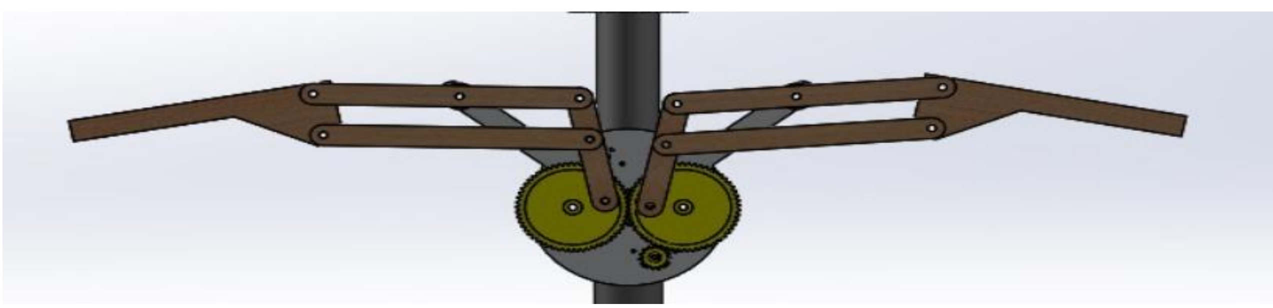
The system will be programmed so that the robot moves along the line, stops at checkpoints, scans the environment, and responds when motion is detected.

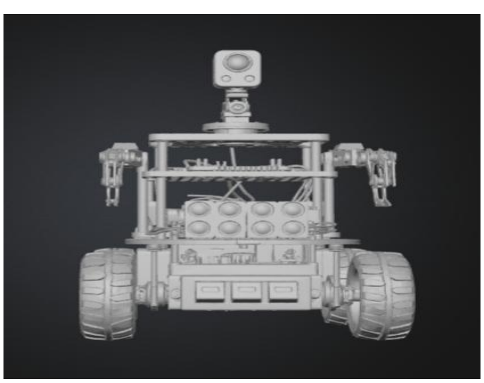
2.5 Design Representation

Block Diagram



Actuator





Chapter 3

Creating a Prototype

A prototype is developed to test whether the proposed robotic scarecrow system functions according to the design. A small-scale version of the robot will be assembled using selected components before building the final system.

The prototype will test the following functions:

- Line-following capability
- Obstacle detection using the ultrasonic sensor
- Stopping at predefined checkpoints
- Rotation of the scanning mechanism at 90° intervals
- Motion detection using the PIR sensor
- Alarm activation when motion is detected
- Actuator movements to scare intruders

This stage helps identify design problems and improve the system before final implementation.

3.1 Prototype Testing

The following tests will be conducted:

- Test line-following sensors to ensure accurate path tracking.
- Test ultrasonic sensor to ensure accurate distance measurement and obstacle detection.
- Verify that the robot stops when an obstacle is detected and continues when it is removed.
- Verify the servo scanning mechanism rotates in 90° intervals.
- Test the PIR sensor detection range.
- Check alarm activation when motion is detected.

3.2 Troubleshooting the Design

Possible issues and solutions include:

- Line detection errors → Adjust sensor calibration.
- Incorrect distance readings → Recheck sensor connections and calibration.
- False PIR detection → Adjust sensor sensitivity.
- Motor instability → Improve power distribution.
- Servo rotation errors → Recheck programming and wiring.

Chapter 4

Building the Robot

After prototype testing, the final robot will be constructed.

Main steps:

1. Assemble the robot chassis.
2. Install DC motors and wheels.
3. Mount line-following sensors at the front.
4. Install the ultrasonic sensor at the front of the robot for obstacle detection.
5. Install the PIR sensor on a rotating servo motor.
6. Connect the buzzer and actuators.
7. Connect the motor driver and Arduino microcontroller.
8. Upload the Arduino program.
9. Test the complete robotic system.

All wiring will be securely connected to ensure safe and reliable operation.

Chapter 5

Evaluating the Robot

5.1 Evaluation of the Design

The robot will be evaluated based on its ability to monitor and protect the garden.

Strengths

- Automatic garden monitoring
- Low-cost implementation
- Improved safety using ultrasonic obstacle detection
- Effective motion detection using PIR sensors
- Simple and easy-to-maintain design

Limitations

- PIR sensor only detects motion
- Limited battery operating time
- More suitable for small garden areas

5.2 Evaluation of the Planning Process

The project will be evaluated based on:

- Completion according to the planned timeline
- Successful prototype testing
- Efficient use of components
- Proper documentation of the development process